Dissertation Report on

Mining Concept-Drift In Big Data Streams

Submitted in partial fulfillment of the requirements for the degree of

Master of Engineering
(Information Technology)

by

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Under the Guidance of

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2012 – 2014

Dedication Sheet

"All Praise, Laud, and Honour to Jesus Christ for his amazing Grace"

Dissertation Approval for M.E.

This dissertation report entitled **"Mining Concept-Drift in Big Data Streams"** by **LIFNA C.S** is approved for the degree of **Master of Engineering** in **Information Technology.**

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Date : 10th September '15

Place : Chembur

Abstract

The digital footprint of the society has undergone an abrupt change due to the extensive use of social networking sites. Analytics upon such social tools help in listening and analyzing customer interactions about products or services and address customer concerns promptly. In today's industry, enterprises are rigorously blending social intelligence with enterprise business intelligence to achieve competitive intelligence. So, the ongoing process of Social Analytics cannot be overlooked. The inferences driven from Social Analytics, are spontaneous and it varies drastically, leading to concept-drift. If used judiciously, Social Analytics can address many social issues as well.

The study was conducted to find concept-drift in Twitter Data Streams. Twitter being an active social tool, rendered exciting leads. The concept-drift was not only established, but there happened to be clues leading to concept-growth as well. The four major domains were identified, as the study was thoroughly conducted to excavate Indian Trends during the Election Process. In contrast to the prevailing state-of-the-art techniques, the proposed system identified eleven other parameters along with frequency, to bring forth accuracy in ranking. Thus, Concept-Drift Identification System (CDIS) was successful in this regard.

The next phase of study was to establish, the temporal relationship between tweets from Human Rights domain. This domain was selected, as it deals with real-time social issues. The NGO from India - Shakti Vahini, which is an authorized organization was zeroed as an authenticated source. When tweets were examined, hidden contexts were deciphered via extensive preprocessing. Three mining techniques were experimented upon the refined dataset to study the confidentiality of the rules. Among them, the performance of the revised Temporal Weighted Association Rule Mining (TWUARM) was magnificent in that process. The study was a great success in regard that, Concept-Drift Mining Engine (CDME) was modeled. In the course of study, transitive property was also revealed. This transitive property was very significant to derive valuable inferences in Human Rights domain to predict events, and take corrective actions. Work is in progress to tune the existing model for the betterment of e-Governance and to incorporate domains other than Human Rights.